

Class Animal

- Variables



```
class Animal {  
    7 usages  
    private static String food;  
    7 usages  
    private static int lifeExpectancy;  
}
```

Class Animal

- Constructors

```
17 // Default constructor
18 4 usages
19 public Animal() {
20     this.food = "unknown";
21     this.lifeExpectancy = 0;
22 }
23
24 // Constructor for food
25 1 usage
26 public Animal(String food) {
27     this.food = food;
28     this.lifeExpectancy = 0;
29 }
30
31 // Constructor for life expectancy
32 1 usage
33 public Animal(int lifeExpectancy) {
34     this.food = "unknown";
35     this.lifeExpectancy = lifeExpectancy;
36 }
37
38 // Constructor for both food and life expectancy
39 1 usage
40 public Animal(String food, int lifeExpectancy) {
41     this.food = food;
42     this.lifeExpectancy = lifeExpectancy;
43 }
```

Class Animal

- Getters and Setters

```
41 // Getters and Setters
42 public String getFood() {
43     return food;
44 }
45
46 public void setFood(String food) {
47     this.food = food;
48 }
49
50 public int getLifeExpectancy() {
51     return lifeExpectancy;
52 }
53
54 public void setLifeExpectancy(int lifeExpectancy) {
55     this.lifeExpectancy = lifeExpectancy;
56 }
57 }
```

Class Animals

- toString();

```
12 // toString method
13 // 1 override
14 public String toString() {
15     return "This " + food + "-eater has a life expectancy of " + lifeExpectancy + " years.";
16 }
```

Class Lion, sub-class (Lion extends Animal)

- Variable

```
8      class Lion extends Animal {  
          10 usages  
9          private static int age;  
          10 usages  
10         private static String name;  
11     }
```

Class Lion, sub-class (Lion extends Animal)

- Constructors

```
45 // Constructor for age
46 // 1 usage
47 public Lion(int age) {
48     super();
49     this.age = age;
50     this.name = "unknown";
51 }
52
53 // Constructor for name
54 // 1 usage
55 public Lion(String name) {
56     super();
57     this.age = 0;
58     this.name = name;
59 }
60
61 // Constructor for food, life expectancy, age, and name
62 // 2 usages
63 public Lion(String food, int lifeExpectancy, int age, String name) {
64     super(food, lifeExpectancy);
65     this.age = age;
66     this.name = name;
67 }
```

```
18 public Lion() {
19     super();
20     this.age = 0;
21     this.name = "unknown";
22 }
23
24 // Constructor for food, age, and name
25 // 1 usage
26 public Lion(String food, int age, String name) {
27     super(food);
28     this.age = age;
29     this.name = name;
30 }
31
32 // Constructor for life expectancy, age, and name
33 // 1 usage
34 public Lion(int lifeExpectancy, int age, String name) {
35     super(lifeExpectancy);
36     this.age = age;
37     this.name = name;
38 }
39
40 // Constructor for age, and name
41 public Lion(int age, String name) {
42     super();
43     this.age = age;
44     this.name = name;
45 }
```

Class Lion, sub-class (Lion extends Animal)

- Getters and Setters

```
66      // Getters and Setters
67      public int getAge() {
68          return age;
69      }
70
71      public void setAge(int age) {
72          this.age = age;
73      }
74
75      public String getName() {
76          return name;
77      }
78
79      public void setName(String name) {
80          this.name = name;
81      }
82  }
```


Class Main

- Variables

```
7 import java.util.Scanner;
8
9 ▶ public class Main {
    2 usages
10     private static String food;
    2 usages
11     private static int lifeExpectation;
    2 usages
12     private static int age;
    2 usages
13     private static String name;
    2 usages
14     private static String input;
    5 usages
15     private static Scanner scanner = new Scanner(System.in);
```

Class Main

- My Lions

```
16  ▶  public static void main(String[] args) {  
17  
18      System.out.println("My Lions:");  
19      Lion myLion = new Lion( food: "meat", lifeExpectancy: 10, age: 5, name: "Leo");  
20      System.out.println(myLion.toString());  
21      Lion myLion1 = new Lion( name: "meat");  
22      System.out.println(myLion1.toString());  
23      Lion myLion2 = new Lion( age: 10);  
24      System.out.println(myLion2.toString());  
25      Lion myLion3 = new Lion( food: "meat", age: 5, name: "Leo");  
26      System.out.println(myLion3.toString());  
27      Lion myLion4 = new Lion( lifeExpectancy: 10, age: 5, name: "Leo");  
28      System.out.println(myLion4.toString());
```

Class Main

- User's Lions

```
30 while (true) {
31     System.out.println("\nWrite something to begin start //'exit' - for exit");
32     input = scanner.next();
33
34     if (input.equalsIgnoreCase( anotherString: "exit")) {
35         break;
36     }
37     System.out.println("Input name: ");
38     name = scanner.next();
39     System.out.println("Input food: ");
40     food = scanner.next();
41     System.out.println("Input lifeExpectation: ");
42     lifeExpectation = scanner.nextInt();
43     System.out.println("Input age: ");
44     age = scanner.nextInt();
45
46     Lion lion = new Lion(food, lifeExpectation, age, name);
47     System.out.println(lion.toString());
48
49     System.out.println("\n\nDo you want to re-run the program?");
50 }
51 }
```

Output

- Testing

```
/Users/dany/Documents/GitHub/projects-in-bife/00P/out/production/00P skillsDemoTwo.Main
```

My Lions:

This meat-eater has a life expectancy of 10 years. Its name is Leo and it is 5 years old.

This unknown-eater has a life expectancy of 0 years. Its name is meat and it is 0 years old.

This unknown-eater has a life expectancy of 0 years. Its name is unknown and it is 10 years old.

This meat-eater has a life expectancy of 0 years. Its name is Leo and it is 5 years old.

This unknown-eater has a life expectancy of 10 years. Its name is Leo and it is 5 years old.

Write something to begin start //'exit' - for exit

start

Input name:

Symbo

Input food:

meat

Input lifeExpectation:

150

Input age:

12

This meat-eater has a life expectancy of 150 years. Its name is Symbo and it is 12 years old.

Do you want to re-run the program?

Write something to begin start //'exit' - for exit

exit

Process finished with exit code 0

- I used my OOP knowledge to create this app. I used the principles of OOP.